

**CH-13 Thermal
Stressors
Homework Answers**



01

Which of the following most influences an employee's ability to dissipate body heat while working in hot environments?

A. Sweat production and evaporation

B. Daily consumption of salt

C. Clothing worn

D. Daily consumption of carbohydrates



02

Calculate the WBGT index with the known information of: outdoor globe temperature of 111 °F (43.9 °C), wet-bulb temperature of 115 °F (46.1 °C), dry-bulb temperature 114 °F (45.6 °C). You will need the following formula: $WBGT = 0.7 WB + 0.2 GT + 0.1 DB$

$$WBGT = 0.7 \times 115 + 0.2 \times 111 + 0.1 \times 114 = 114.1$$

A. 101.3 °F (38.5 °C)

B. 115 °F (46.1 °C)

C. 114.1 °F (45.6 °C)

D. 112.1 °F (44.5 °C)



03

Which of the following is an appropriate control for heat stress?

- A. Encourage workers to drink small amounts of water about every 20 minutes
- B. Allow employees to self-limit exposure to heat
- C. Remove workers from hot environments when core body temperatures exceed 38.5oC
- D. All of the above are appropriate heat stress controls**



04

Select the correct definition for the heat stress index.

- A. The ratio of relative humidity to absolute humidity.
- B. The ratio of wet bulb temperature to dry bulb temperature.
- C. The ratio of the evaporative heat loss required to maintain body temperature to the maximum evaporative heat loss.**
- D. The ratio of the dew point to 50% relative humidity.



05

Which set of signs and symptoms is characteristic of a worker experiencing heat stroke?

A. Skin eruptions, itching red skin, reduced sweating

B. Chills, restlessness, mental confusion, hot dry skin

C. Profuse sweating, vomiting, high pulse rate, blurred vision

D. Incapacitating muscular cramping in abdominal and fatigued muscle groups



06

Administrative work practice controls for heat stress include acclimatization. This requires exposing the worker to the work environment for how long?

A. Two hours per day for 1 to 2 weeks

B. Four hours per day for 1 week

C. One full week

D. Two full days



07

The ACGIH has established cold stress TLV's for workers in cold occupations. At what body core temperature (°F) is the worker in danger of developing severe hypothermia if his/her core temperature dropped further?

A. 75.5°F

B. 80.4°F

C. 85.8°F

D. 91.4°F



08

Hypothermia is defined as a condition when:

- A. Death occurs
- B. Extreme high blood pressure
- C. The body gains heat faster than it can lose it
- D. The body loses heat faster than it can produce it**



09

Industrial hygienists MOST commonly measure thermal conditions to control employee thermal stress using the wet bulb globe temperature (WBGT) index and which another Index?

A. Temperature/Humidity/Sun/Wind (THSW) index

B. Apparent Temperature (AT) index

C. Temperature Humidity (THI) index

D. Wind Chill (WC) index



10

Welder is on a ship deck welding steel plates to the interior of the ship. The temperature in the vault is 95 degrees with the humidity of 85% after two hours of

operation the welder complaints he doesn't feel good. He has pale white complexion, nausea, headaches, rapid pulse and low blood pressure.

Which describes the condition that most likely caused his illness?

A. Siderosis

B. Welder Flash Syndrome

C. Heat exhaustion

D. Chrome Fume fever